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Problem Statement:

As presented by Google, determining whether a search result is malicious or safe, or even whether it is a sponsored ad or an organic search result can be very difficult. This problem is being abused by threat actors to spread malware.

Motivation:

One of the most successful tactics of threat actors I have observed in my work as a threat hunter, is the creation of malicious websites to distribute malware, and then advertising those sites via malvertising, or paying Google to present these sites in search results. The way these ads are displayed makes them very difficult for less technically educated users to distinguish them from actual results. Due to this, many users, when presented with a sponsored malicious search result believe it to be an actual safe result that meets the requirements of what they were looking for with their search.

I was inspired to tackle this problem by my own mother, who was nearly a victim of one of these scams. She had been fooled by one of these links and had nearly downloaded a piece of malicious software but decided to double check with me before she did. We discovered that she was not at the legitimate site she thought she was on.

The most obvious solution to this problem would be a Firefox add-on that could review a link before my mother clicked it. If suspicious, the add-on would make the link background red, or in some other way, identify to the user that the link is suspicious.

I searched for existing products meeting this need and did not find any maintained add-ons that fill this specific niche. There are ad and scam blocking add-ons, however these attempt to remove the results entirely, and also work as ad-blockers. While useful, there are many cases where a user may want to still see all the content presented, but to have more info about each link.

I also found a few projects that seem to now be defunct. Web of Trust, which hasn't been updated in three years, and has a homepage that no longer resolves, claims to solve a similar problem, but seems to be no longer maintained. Malicious URL Detector sounds exactly like what I'm proposing, but it hasn't been updated in 7 years. Scam Site Detector sounds similar as well, but it has 0 reviews and only 1 user, likely the developer. These are the best matches I could find up to this point.

Expected Deliverables:

By the time I am complete, I expect to have a fully functioning and working Firefox add-on that will examine Google Search Results. It will look at the sponsored results and if any are suspicious, it will make the background of them red, or create a pop-up asking for confirmation before allowing the user to navigate to the site, or in some other way, warn the user that the URL is not trustworthy.

The above is my minimum expected deliverable, and I am confident I can achieve this in the time-frames posted. As I do not work as a developer professionally, I am not yet clear on how long this will take to accomplish. Fortunately, this problem lends itself easily to further work via additional features. If I reach my original goal early, I can add the ability to use other non-google search engines, the ways malicious links are identified could be expanded (non-ascii characters, newly created sites, non-english comments in english presenting sites, typo-squatting, comparison against open source known malicious domain lists, machine learning, LLM powered explanations, etc.).

Links and References:

My WOT: <https://addons.mozilla.org/en-US/firefox/addon/wot-safe-browsing-tool/>

Malicious URL Detector: <https://addons.mozilla.org/en-US/firefox/addon/malicious-url-detector/>

Scam Site Detector: <https://addons.mozilla.org/en-US/firefox/addon/scam-site-detector/>